Polynomial Operations

Concept

Polynomials combine terms with non-negative integer variable exponents.

Degree

The highest variable exponent after simplification.

Like terms

Same variable part; add or subtract their coefficients.

Subtract

Distribute the minus sign to every term in the bracket.

Distribute

Multiply or divide every term by the monomial; keep restrictions.

Example 1

Simplify $(5x - 2) - (3x + 8)$.

Solution

1. Remove brackets: $5x - 2 - 3x - 8$.

2. Collect like terms: $2x - 10$.

Polynomial Operations

Example 2

Expand $-2x(3x - 4)$.

Solution

1. Multiply each term: $(-2x)(3x) + (-2x)(-4)$.
2. Result: $-6x^2 + 8x$.

Example 3

Simplify $(12x^2 - 18x)/(6x)$.

Solution

1. $x \neq 0$. Divide each term by $6x$.
2. Result: $2x - 3$, with $x \neq 0$ retained.

Polynomial Operations

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. State degree and number of terms of $7 - 2x^2 + x$.

2. Simplify $(4x^2 - x + 2) - (x^2 + 3x - 5)$.

3. Expand $3x(2x + 5)$.

4. Simplify $(15x^2 + 10x)/(5x)$.

5. Explain why $2x + 3x^2$ is not $5x^3$.

Polynomial Operations

Practice answers | check after trying

1. Degree 2; three terms, so a trinomial.

2. $3x^2 - 4x + 7$.

3. $6x^2 + 15x$.

4. $3x + 2$, $x \neq 0$.

5. The terms are unlike and are being added, not multiplied. At $x = 2$ the original is 16 while $5x^3$ is 40.